

Q3 - Student Registration Form

Java Source Code:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class StudentRegistration extends JFrame {

    JTextField idField, nameField;
    JRadioButton male, female;
    JComboBox<String> department;
    JButton register, clear;
    JTextArea result;

    StudentRegistration() {
        setTitle("Student Registration Form");
        setSize(500, 500);
        setLayout(new FlowLayout());

        add(new JLabel("Student ID:"));
        idField = new JTextField(20);
        add(idField);

        add(new JLabel("Student Name:"));
        nameField = new JTextField(20);
        add(nameField);

        add(new JLabel("Gender:"));
        male = new JRadioButton("Male");
        female = new JRadioButton("Female");
        ButtonGroup genderGroup = new ButtonGroup();
        genderGroup.add(male);
        genderGroup.add(female);
        add(male);
        add(female);

        add(new JLabel("Department:"));
        String[] departments = {"Information Technology", "Computer Science",
                                "Electronics", "Mechanical"};
        department = new JComboBox<>(departments);
        add(department);

        register = new JButton("Register");
        clear = new JButton("Clear");
        add(register);
        add(clear);

        result = new JTextArea(10, 35);
        add(result);
    }
}
```

```

        register.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                String gender = "";
                if (male.isSelected()) gender = "Male";
                else if (female.isSelected()) gender = "Female";

                result.setText(
                    "Student Registered Successfully
" +
                    "Student ID: " + idField.getText() + "
" +
                    "Student Name: " + nameField.getText() + "
" +
                    "Gender: " + gender + "
" +
                    "Department: " + department.getSelectedItem()
                );
            }
        });

        clear.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                idField.setText("");
                nameField.setText("");
                genderGroup.clearSelection();
                department.setSelectedIndex(0);
                result.setText("");
            }
        });

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);
    }

    public static void main(String[] args) {
        new StudentRegistration();
    }
}

```

Compile / Run:

```

javac StudentRegistration.java
java StudentRegistration

```

Expected Output:

```

Student Registered Successfully
Student ID: S1012
Student Name: Priya
Gender: Male
Department: Information Technology

```